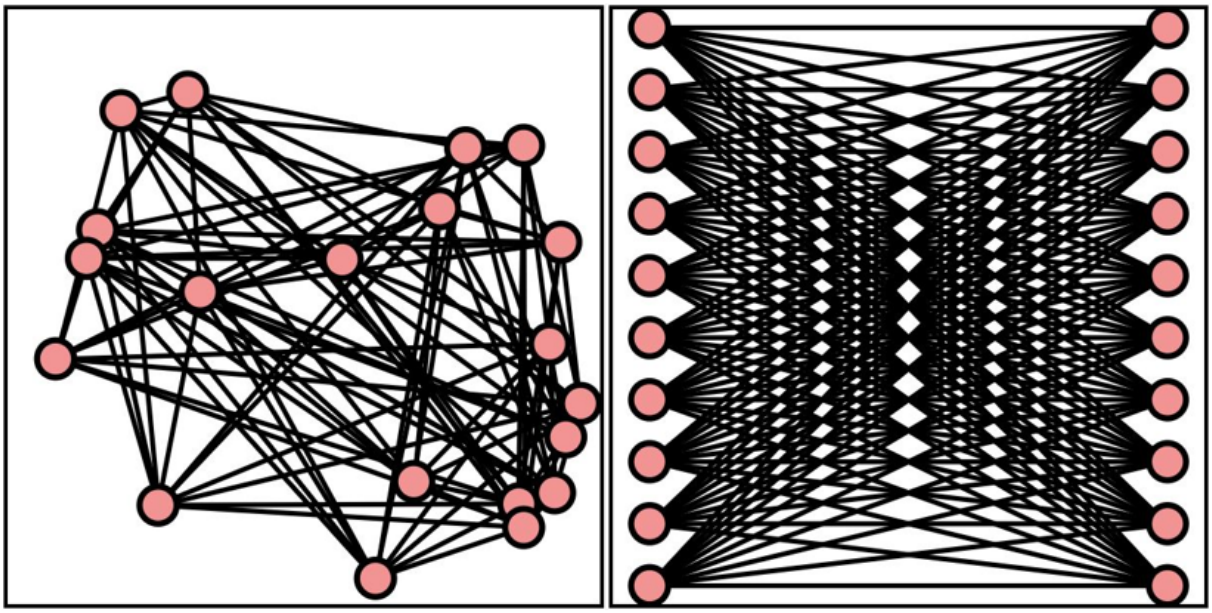


Figure 17.16 A nice embedding can make the difference in understanding a graph: complete bipartite graph $K_{10,10}$ is drawn randomly (left) and symmetrically (right): which one shows the structure of this graph better?

To no surprise, simulated annealing can help in this case as well: Davidson and Harel first used this technique[16] to converge to local optima while staying out of local pitfalls.

The problem with using standard simulated annealing is that random search makes convergence too slow; to work around this limitation, several authors suggested using hybrid solutions that leverage the strengths of both approaches. GEM[17] algorithm stands among them for its innovative approach and impressive results.

GEM is not using simulated annealing, but it borrows the concept of temperature from it: there is no cooling cycle, rater the temperate (which is still expressing the degree of